

Gamma Distribution

A continuous r.v X is said to follow Gamma distribution if the pdf of X is as follows:

$$f(x) = \begin{cases} \frac{e^{-x} x^{\lambda-1}}{\Gamma(\lambda)}; \lambda > 0, 0 < x < \infty \\ 0, \text{ otherwise} \end{cases}$$

here λ is the parameter of the distribution, X is called $\gamma(\lambda)$ variate and denoted as $X \sim \gamma(\lambda)$
Its a single parameter Gamma distribution. Now the pdf can be as follows:

$$f(x) = \begin{cases} \frac{a^\lambda e^{-ax} x^{\lambda-1}}{\Gamma(\lambda)}; a > 0, \lambda > 0, 0 < x < \infty \\ 0, \text{ otherwise} \end{cases}$$

the X is said to follow a two parameter Gamma distribution and X is called two parameter Gamma variate where a, λ are the two parameters., it is denoted as $X \sim \gamma(a, \lambda)$ when $a=1$ we get the above mentioned single parameter Gamma distribution.

$$\text{CDF of Gamma distribution is } F(x) = \begin{cases} \int_0^x f(x) dx = \int_0^x \frac{e^{-x} x^{\lambda-1}}{\Gamma(\lambda)} dx \\ 0, \text{ otherwise} \end{cases}$$

Additive Property of Gamma Distribution

If $X_1, X_2, \dots, X_n$ are independent Gamma variates with parameters $\lambda_1, \lambda_2, \dots, \lambda_n$ respectively then $X_1 + X_2 + X_3 + \dots + X_n$ will follow Gamma distribution with parameter $\lambda_1, \lambda_2, \dots, \lambda_n$.

Moment Generating Function of Gamma Distribution:

The mgf is given by

$$M_x(t) = \frac{1}{\Gamma(\lambda)} \int_0^\infty e^{tx} e^{-x} x^{\lambda-1} dx = \frac{1}{\Gamma(\lambda)} \int_0^\infty e^{-(1-t)x} x^{\lambda-1} dx = \frac{1}{\Gamma(\lambda)} \times \frac{\Gamma(\lambda)}{(1-t)^\lambda} = (1-t)^{-\lambda}, |t| < 1$$

Mean and Variance of Gamma distribution using Gamma Distribution using mgf:

Mean = $\mu_x = E(X) = M'_x(t)$ at $t=0 = \lambda(1-t)^{-\lambda-1}$ at $t=0 = \lambda$ and variance=

$$\sigma^2 = E(X^2) - \{E(X)\}^2 = M''_x(t) - \{M'_x(t)\}^2 \text{ at } t=0 = \lambda(\lambda+1)(1-t)^{-\lambda-2} - \lambda^2 \text{ at } t=0 = \lambda^2 + \lambda - \lambda^2 = \lambda$$

Hence, like Poisson distribution the mean and variance of Gamma distribution are also equal.

When $\lambda \rightarrow \infty$ Gamma distribution tends to Standard Normal distribution. This is known as limiting form of Gamma distribution.

Example

The daily consumption of milk in city in excess of 20,000 gallons, is approximately distributed as Gamma variate with parameters $a = 2, \lambda = \frac{1}{10,000}$. The city has daily stock of 30,000 gallons. What is the probability that the stock is insufficient on a particular day.

Ans: Let X is the r.v denotes the daily consumption of milk in gallons in the city,

$$\text{Then Prob.}(X > 30,000) = \int_{30,000}^{\infty} f(x) dx = \int_{30,000}^{\infty} \frac{e^{-x} x^{\lambda-1}}{\Gamma(\lambda)} dx = \frac{2}{e}$$